

# Chapter 4

## Multivariate Random Variables

In this chapter, we will review some topics related to random vectors, which will be of use in the following chapters.

### 4.1 Gaussian Random Vectors (Multivariate Normal Distribution)

Recall that a random variable  $X$  is Gaussian (normal) with mean  $\mu$  and variance  $\sigma^2 > 0$  if the pdf of  $X$  is given by

$$p_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp -\frac{(x - \mu)^2}{2\sigma^2}. \quad (4.1)$$

**Definition 4.1.** A collection of random variables is **jointly Gaussian** if any linear combination of these variables is Gaussian. A **Gaussian random vector**, also known as a *multivariate normal vector*, is a vector whose elements are jointly Gaussian. A collection of random vectors is jointly Gaussian if the vector obtained by concatenating them is jointly Gaussian.

**Example 4.2.** If  $\begin{pmatrix} X \\ Y \end{pmatrix}$  is a Gaussian vector, then  $Z = 2X + 3Y$  is Gaussian. Furthermore,

$$\mathbb{E}[Z] = 2\mathbb{E}[X] + 3\mathbb{E}[Y], \quad (4.2)$$

$$\text{Var}(Z) = \text{Cov}(2X + 3Y, 2X + 3Y) = 4\text{Cov}(X, X) + 12\text{Cov}(X, Y) + 9\text{Cov}(Y, Y) \quad (4.3)$$

$$= 4\text{Var}(X) + 12\text{Cov}(X, Y) + 9\text{Var}(Y), \quad (4.4)$$

which completely characterizes the distribution of  $Z$  as  $Z \sim \mathcal{N}(\mathbb{E}[Z], \text{Var}(Z))$ .  $\triangle$

For a Gaussian random vector  $\mathbf{X}$  of dimension  $d$ , with mean  $\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu}$  and covariance matrix  $\mathbf{K} = \text{Cov}(\mathbf{X}) = \mathbb{E}[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$ , we have

$$p_{\mathbf{X}}(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} |\mathbf{K}|^{1/2}} \exp \left( -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^T \mathbf{K}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right), \quad (4.5)$$

provided that the covariance matrix is invertible.

The elements of  $\mathbf{X}$  are **independent** if and only if the covariance matrix is diagonal.

#### 4.1.1 Maximum likelihood estimation

Consider a  $d$ -dimensional random vector  $\mathbf{X} = (X_1, \dots, X_d)$  with distribution  $\mathcal{N}(\boldsymbol{\theta}^*, \mathbf{K}^*)$  given in (4.5), where  $\boldsymbol{\theta}^*, \mathbf{K}^*$  are unknown. Suppose we are interested in the relationship between  $X_d$  and  $X_1, \dots, X_{d-1}$ . For

example, for  $\mathbf{X}^T = (X_1, X_2, X_3)$ ,  $X_1$  and  $X_2$  could indicate the heights of the parents and  $X_3$  could be the height of the child. We may, for example, be interested in finding  $\mathbb{E}[X_d | X_1, \dots, X_{d-1}]$ , thus estimating  $X_d$  based on  $X_1, \dots, X_{d-1}$ . If we find the distribution, in other words,  $\boldsymbol{\theta}^*, \mathbf{K}^*$ , we can do so. Furthermore, the matrix  $\mathbf{K}^*$  can indicate which dimensions are more strongly correlated.

Consider a set of  $n$  iid samples  $\mathcal{D} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ , where each  $\mathbf{x}_i$  is a sample of  $\mathbf{X}$ . We denote the elements of  $\mathbf{x}_i$  as  $\mathbf{x}_i = (x_{i1}, \dots, x_{id})$ .

To estimate  $\boldsymbol{\theta}^*$  and  $\mathbf{K}^*$ , we write

$$\ell(\boldsymbol{\theta}, \mathbf{K}) = \ln p(\mathcal{D}; \boldsymbol{\theta}, \mathbf{K}) = \sum_{i=1}^n \ln p(\mathbf{x}_i; \boldsymbol{\theta}, \mathbf{K}) \quad (4.6)$$

$$\doteq \frac{n}{2} \ln |\mathbf{K}^{-1}| - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{K}^{-1} (\mathbf{x}_i - \boldsymbol{\theta}), \quad (4.7)$$

where we have used the fact that  $|\mathbf{K}^{-1}| = \frac{1}{|\mathbf{K}|}$ .

As seen in the appendix (last chapter), for a symmetric matrix  $\mathbf{A}$ , we have  $\frac{d}{dv}(\mathbf{y}^T \mathbf{A} \mathbf{y}) = 2\mathbf{y}^T \mathbf{A} \frac{d\mathbf{y}}{dv}$ . Hence,

$$\frac{\partial \ell}{\partial \boldsymbol{\theta}} = -\frac{1}{2} \sum_{i=1}^n 2(\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{K}^{-1} (-\mathbf{I}) = \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{K}^{-1}. \quad (4.8)$$

Setting this equal to zero yields

$$\hat{\boldsymbol{\theta}}_{ML} = \bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i. \quad (4.9)$$

**Exercise 4.3.** Using the facts

$$\frac{\partial}{\partial \mathbf{A}} \mathbf{x}^T \mathbf{A} \mathbf{x} = \mathbf{x} \mathbf{x}^T, \quad \frac{\partial}{\partial \mathbf{A}} \ln |\mathbf{A}| = \mathbf{A}^{-T} \quad (4.10)$$

prove that

$$\hat{\mathbf{K}}_{ML} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T \quad (4.11)$$

△

### 4.1.2 Bayesian estimation

We now solve the same problem using Bayesian estimation, with the following likelihood

$$\mathbf{X} | \boldsymbol{\Theta} \sim \mathcal{N}(\boldsymbol{\Theta}, \mathbf{K}), \quad (4.12)$$

$$p(\mathbf{x}_1^n | \boldsymbol{\theta}) \propto \exp \left( -\frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{K}^{-1} (\mathbf{x}_i - \boldsymbol{\theta}) \right), \quad (4.13)$$

where, for simplicity, we assume  $\mathbf{K}$  is known and we only need to estimate  $\boldsymbol{\Theta}$ . As the prior, we choose

$$\boldsymbol{\Theta} \sim \mathcal{N}(\boldsymbol{\mu}_0, \mathbf{S}_0) \quad (4.14)$$

$$p(\boldsymbol{\theta}) \propto \exp \left( -\frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \mathbf{S}_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) \right). \quad (4.15)$$

Hence,

$$p(\boldsymbol{\theta} | \mathbf{x}_1^n) \propto \exp \left( -\frac{1}{2} (\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \mathbf{S}_0^{-1} (\boldsymbol{\theta} - \boldsymbol{\mu}_0) - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\theta})^T \mathbf{K}^{-1} (\mathbf{x}_i - \boldsymbol{\theta}) \right). \quad (4.16)$$

The exponent in the posterior is quadratic in  $\boldsymbol{\theta}$ , indicating that  $\boldsymbol{\Theta}$  has a Gaussian distribution. So  $\boldsymbol{\Theta}|\mathbf{x}_1^n \sim \mathcal{N}(\hat{\boldsymbol{\theta}}_n, \mathbf{S}_n)$ , for appropriate choices of  $\hat{\boldsymbol{\theta}}_n$  and  $\mathbf{S}_n$ ,

$$p(\boldsymbol{\theta}|\mathbf{x}_1^n) \propto \exp\left(-\frac{1}{2}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n)^T \mathbf{S}_n^{-1}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n)\right). \quad (4.17)$$

To find  $\hat{\boldsymbol{\theta}}_n$  and  $\mathbf{S}_n$ , we equate (4.16) and (4.17), ignoring constant multiplicative factors, which leads to

$$(\boldsymbol{\theta} - \boldsymbol{\mu}_0)^T \mathbf{S}_0^{-1}(\boldsymbol{\theta} - \boldsymbol{\mu}_0) + \sum_{i=1}^n (\boldsymbol{\theta} - \mathbf{x}_i)^T \mathbf{K}^{-1}(\boldsymbol{\theta} - \mathbf{x}_i) \quad \doteq \quad (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n)^T \mathbf{S}_n^{-1}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_n), \quad (4.18)$$

$$\boldsymbol{\theta}^T \mathbf{S}_0^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathbf{S}_0^{-1} \boldsymbol{\mu}_0 + n\boldsymbol{\theta}^T \mathbf{K}^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathbf{K}^{-1} \sum_{i=1}^n \mathbf{x}_i \quad \doteq \quad \boldsymbol{\theta}^T \mathbf{S}_n^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathbf{S}_n^{-1} \hat{\boldsymbol{\theta}}_n. \quad (4.19)$$

Here, we have used the fact that

$$(\mathbf{a} - \mathbf{b})^T \mathbf{A}(\mathbf{a} - \mathbf{b}) = \mathbf{a}^T \mathbf{A} \mathbf{a} - \mathbf{a}^T \mathbf{A} \mathbf{b} - \mathbf{b}^T \mathbf{A} \mathbf{a} + \mathbf{b}^T \mathbf{A} \mathbf{b} = \mathbf{a}^T \mathbf{A} \mathbf{a} - 2\mathbf{a}^T \mathbf{A} \mathbf{b} + \mathbf{b}^T \mathbf{A} \mathbf{b},$$

for vectors  $\mathbf{a}, \mathbf{b}$  and a symmetric matrix  $\mathbf{A}$ . Note that  $\mathbf{a}^T \mathbf{A} \mathbf{b} = \mathbf{b}^T \mathbf{A} \mathbf{a}$ , as both sides are scalars and  $\mathbf{a}^T \mathbf{A} \mathbf{b} = (\mathbf{a}^T \mathbf{A} \mathbf{b})^T = \mathbf{b}^T \mathbf{A} \mathbf{a}$ .

We now collect the terms of the form  $\boldsymbol{\theta}^T \mathbf{A} \boldsymbol{\theta}$ ,

$$\boldsymbol{\theta}^T (\mathbf{S}_0^{-1} + n\mathbf{K}^{-1}) \boldsymbol{\theta} - 2\boldsymbol{\theta}^T (\mathbf{S}_0^{-1} \boldsymbol{\mu}_0 + \mathbf{K}^{-1} \sum_{i=1}^n \mathbf{x}_i) \doteq \boldsymbol{\theta}^T \mathbf{S}_n^{-1} \boldsymbol{\theta} - 2\boldsymbol{\theta}^T \mathbf{S}_n^{-1} \hat{\boldsymbol{\theta}}_n, \quad (4.20)$$

leading to the following values for the parameters of the posterior distribution  $\boldsymbol{\Theta}|\mathbf{x}_1^n \sim \mathcal{N}(\hat{\boldsymbol{\theta}}_n, \mathbf{S}_n)$ ,

$$\mathbf{S}_n^{-1} = \mathbf{S}_0^{-1} + n\mathbf{K}^{-1}, \quad (4.21)$$

$$\hat{\boldsymbol{\theta}}_n = \mathbf{S}_n (\mathbf{S}_0^{-1} \boldsymbol{\mu}_0 + n\mathbf{K}^{-1} \bar{\mathbf{x}}) \quad (4.22)$$

$$= (\mathbf{S}_0^{-1} + n\mathbf{K}^{-1})^{-1} (\mathbf{S}_0^{-1} \boldsymbol{\mu}_0 + n\mathbf{K}^{-1} \bar{\mathbf{x}}), \quad (4.23)$$

where  $\bar{\mathbf{x}}$  is  $\sum_{i=1}^n \mathbf{x}_i / n$ . The posterior mean,  $\hat{\boldsymbol{\theta}}_n$ , which we can also view as a point estimate, is the weighted average of the prior mean  $\boldsymbol{\mu}_0$  and what is suggested by the data  $\bar{\mathbf{x}}$ .

**Exercise 4.4.** Find  $\hat{\boldsymbol{\theta}}_n$  and  $\mathbf{S}_n^{-1}$  when  $\mathbf{S}_0 = s^2 \mathbf{I}$  and  $\mathbf{K} = \sigma^2 \mathbf{I}$  and interpret the results.  $\triangle$